Assignment 2

Part I

1. (a) Suppose

$$f(n_1, n_2) = e^{-\lambda}\frac{\lambda^{n_1}}{n_1!}\binom{n_1}{n_2}p^{n_2}(1-p)^{n_1-n_2}, \qquad 0 \le n_2 \le n_1.$$

Then

$$\begin{aligned} f_{X_1}(n_1) &= \sum_{n_2=0}^{n_1} f(n_1, n_2) \\ &= \sum_{n_2=0}^{n_1} e^{-\lambda}\frac{\lambda^{n_1}}{n_1!}\binom{n_1}{n_2}p^{n_2}(1-p)^{n_1-n_2} \\ &= e^{-\lambda}\frac{\lambda^{n_1}}{n_1!}\sum_{n_2=0}^{n_1}\binom{n_1}{n_2}p^{n_2}(1-p)^{n_1-n_2}. \end{aligned}$$

By the binomial theorem,

$$\sum_{n_2=0}^{n_1}\binom{n_1}{n_2}p^{n_2}(1-p)^{n_1-n_2} = (p+1-p)^{n_1} = 1.$$

Therefore,

$$f_{X_1}(n_1) = e^{-\lambda}\frac{\lambda^{n_1}}{n_1!},$$

so

$$X_1 \sim \text{Poisson}(\lambda).$$

Also,

$$\begin{aligned} f_{X_2}(n_2) &= \sum_{n_1=n_2}^{\infty} f(n_1, n_2) \\ &= \sum_{n_1=n_2}^{\infty} e^{-\lambda}\frac{\lambda^{n_1}}{n_1!}\frac{n_1!}{n_2!(n_1-n_2)!}p^{n_2}(1-p)^{n_1-n_2} \\ &= e^{-\lambda}\frac{(\lambda p)^{n_2}}{n_2!}\sum_{n_1=n_2}^{\infty}\frac{(\lambda(1-p))^{n_1-n_2}}{(n_1-n_2)!}. \end{aligned}$$

Let $k = n_1 - n_2$, so $n_1 = k + n_2$. Then

$$\begin{aligned} f_{X_2}(n_2) &= e^{-\lambda} \frac{(\lambda p)^{n_2}}{n_2!} \sum_{k=0}^{\infty} \frac{(\lambda(1-p))^k}{k!} \\ &= e^{-\lambda} \frac{(\lambda p)^{n_2}}{n_2!} e^{\lambda(1-p)} \\ &= e^{-\lambda p} \frac{(\lambda p)^{n_2}}{n_2!}. \end{aligned}$$

Thus

$$X_2 \sim \text{Poisson}(\lambda p).$$

(b) Let

$$Y = X_1 - X_2, \qquad X_1 = X_2 + Y.$$

The joint PMF of (X_2, Y) is

$$f_{X_2,Y}(n_2, y) = f(n_2 + y, n_2).$$

Thus

$$\begin{aligned} f_{X_2,Y}(n_2, y) &= e^{-\lambda} \frac{\lambda^{n_2+y}}{(n_2+y)!} \binom{n_2+y}{n_2} p^{n_2} (1-p)^y \\ &= e^{-\lambda} \frac{\lambda^{n_2+y}}{n_2! y!} p^{n_2} (1-p)^y \\ &= \left(e^{-\lambda p} \frac{(\lambda p)^{n_2}}{n_2!} \right) \left(e^{-\lambda(1-p)} \frac{(\lambda(1-p))^y}{y!} \right) \\ &= f_{X_2}(n_2) f_Y(y). \end{aligned}$$

Therefore,

$$X_2 \sim \text{Poisson}(\lambda p), \qquad Y \sim \text{Poisson}(\lambda(1-p)),$$

and X_2 and Y are independent.

2. Let

$$P(\text{false coin}) = \frac{m}{n}, \qquad P(\text{true coin}) = 1 - \frac{m}{n}.$$

For each coin i, define

$$X_i = \begin{cases} 1, & \text{if the selected coin is false,} \\ 0, & \text{if the selected coin is true.} \end{cases}$$

Then

$$P(X_i = 1) = \frac{m}{n}, \qquad P(X_i = 0) = 1 - \frac{m}{n},$$

so

$$X_i \sim \text{Bernoulli}\left(\frac{m}{n}\right).$$

Let X denote the total number of false coins:

$$X = \sum_{i=1}^{n} X_i.$$

Then

$$X \sim \text{Binomial}\left(n, \frac{m}{n}\right),$$

and

$$P(X = r) = \binom{n}{r} \left(\frac{m}{n}\right)^r \left(1 - \frac{m}{n}\right)^{n-r}.$$

As $n \to \infty$,

$$p = \frac{m}{n} \to 0, \qquad np = m.$$

Therefore,

$$\text{Binomial}\left(n, \frac{m}{n}\right) \to \text{Poisson}(m),$$

and hence

$$P(X = r) = e^{-m} \frac{m^r}{r!}.$$

3. (a) The normalizing constant c satisfies

$$\int_0^\infty \int_0^{n_1} ce^{-n_1}\, dn_2\, dn_1 = 1.$$

Thus

$$\begin{aligned}
\int_0^\infty cn_1 e^{-n_1}\, dn_1 &= 1, \\
\int_0^\infty n_1 e^{-n_1}\, dn_1 &= \left[-n_1 e^{-n_1}\right]_0^\infty - \int_0^\infty (-e^{-n_1})\, dn_1 \\
&= \left[-e^{-n_1}\right]_0^\infty = 1.
\end{aligned}$$

Therefore,

$$c = 1.$$

(b) The conditional density is

$$f_{X_2|X_1}(n_2 \mid n_1) = \frac{f_{X_1,X_2}(n_1, n_2)}{f_{X_1}(n_1)}.$$

Now

$$f_{X_1}(n_1) = \int_0^{n_1} e^{-n_1}\, dn_2 = n_1 e^{-n_1}, \qquad n_1 \geq 0.$$

Hence

$$f_{X_2|X_1}(n_2 \mid n_1) = \begin{cases} \dfrac{1}{n_1}, & 0 \leq n_2 \leq n_1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) We need to find $P(X_2 > 1 \mid X_1 = 2)$. Since

$$f_{X_2|X_1}(n_2 \mid 2) = \begin{cases} \dfrac{1}{2}, & 0 \leq n_2 \leq 2, \\ 0, & \text{otherwise,} \end{cases}$$

we get

$$P(X_2 > 1 \mid X_1 = 2) = \int_1^2 \frac{1}{2}\, dn_2 = \frac{1}{2}.$$

4. Let

$$P(N = n) = e^{-\lambda}\frac{\lambda^n}{n!},$$

and suppose

$$E(X_i) = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2.$$

Let

$$S = \sum_{i=1}^{N} X_i,$$

with $S = 0$ when $N = 0$.

By the law of total expectation,

$$E(S) = E\big(E(S \mid N)\big).$$

Since

$$E(S \mid N = n) = E(X_1 + X_2 + \cdots + X_n) = n\mu,$$

we have

$$E(S \mid N) = N\mu.$$

Therefore,

$$E(S) = E(N\mu) = \mu E(N) = \mu\lambda.$$

By the law of total variance,

$$\mathrm{Var}(S) = E\big(\mathrm{Var}(S \mid N)\big) + \mathrm{Var}\big(E(S \mid N)\big).$$

Now

$$\mathrm{Var}(S \mid N = n) = n\sigma^2, \qquad \mathrm{Var}(S \mid N) = N\sigma^2.$$

Thus

$$E\big(\mathrm{Var}(S \mid N)\big) = \sigma^2 E(N) = \sigma^2\lambda.$$

Also,

$$\mathrm{Var}\big(E(S \mid N)\big) = \mathrm{Var}(N\mu) = \mu^2\,\mathrm{Var}(N) = \mu^2\lambda.$$

Hence

$$\mathrm{Var}(S) = \sigma^2\lambda + \mu^2\lambda = \lambda(\sigma^2 + \mu^2).$$

5. The covariance formula is

$$\mathrm{Cov}(X,Y) = E\big((X-E(X))(Y-E(Y))\big) = E(XY) - E(X)E(Y).$$

Two random variables X and Y are uncorrelated if

$$\mathrm{Cov}(X,Y) = 0.$$

(a) Let

$$U = X_1 + X_2, \qquad V = X_1 - X_2.$$

Then

$$\begin{aligned}\mathrm{Cov}(U,V) &= \mathrm{Cov}(X_1+X_2, X_1-X_2)\\ &= \mathrm{Cov}(X_1,X_1) - \mathrm{Cov}(X_2,X_2)\\ &= \mathrm{Var}(X_1) - \mathrm{Var}(X_2)\\ &= 0.\end{aligned}$$

Thus U and V are uncorrelated.

(b) Let $Y = X^2$, where

$$X \in \{-1, 0, 1\}.$$

Then

$$\begin{aligned}\mathrm{Cov}(X,X^2) &= E(X^3) - E(X)E(X^2)\\ &= 0 - 0\\ &= 0.\end{aligned}$$

So X and X^2 are uncorrelated, but X^2 is dependent on X.

6. Let urn 1 and urn 2 contain T balls. Let urn 1 contain w_1 white balls and urn 2 contain w_2 white balls. Define

$$p = P(\text{white from urn 1}) = \frac{w_1}{T}, \qquad q = P(\text{white from urn 2}) = \frac{w_2}{T}.$$

We want a value of p such that

$$p^n = q^n + (1-q)^n, \qquad 0 \le p, q \le 1.$$

Since $0 \le q \le 1$, we have $0 \le 1-q \le 1$. For $n \ge 1$,

$$q^n \le q, \qquad (1-q)^n \le 1-q.$$

Therefore,

$$q^n + (1-q)^n \le q + (1-q) = 1.$$

Thus, for all $q \in [0,1]$, there exists a p such that

$$p = \left(q^n + (1-q)^n\right)^{1/n} \le 1.$$